



Experimental study on Na⁺ conductivity in NaAlBr₄ and atomic-scale investigation of Na⁺ conduction

Reona Miyazaki¹ · Masanobu Nakayama² · Takehiko Hihara¹

Received: 1 August 2024 / Revised: 10 September 2024 / Accepted: 11 September 2024 / Published online: 20 September 2024
© The Author(s) 2024

Abstract

The ionic conduction properties of Li/Na metal halides have been extensively studied, with recent attention turning towards Al-based systems. However, limited studies have focused on alkali Al bromides. In this study, we explored the Na⁺ conduction properties of NaAlBr₄. Conductivity measurements at 30 °C revealed a Na⁺ conductivity of 1.2×10^{-5} S/cm, surpassing that of isostructural NaAlCl₄ threefold. Molecular dynamics (MD) simulations to elucidate the conduction mechanisms revealed that Na⁺ conduction was not observed in stoichiometric NaAlBr₄, which has high formation energies of Na⁺ vacancies and interstitials (0.88 eV and 0.73 eV, respectively). Nevertheless, a conductivity of 1.2×10^{-5} S/cm was observed. The activation energy for ion conduction was experimentally determined as 0.43 eV, and the migration energies were calculated as 0.26 eV (Na⁺ vacancies) and 0.16 eV (Na⁺ interstitials) by MD simulations. These discrepancies in ion conduction were partially explained by the role of transient defects enriched via ball milling in facilitating Na⁺ conduction on the particle surface, offering insights into the complex ion conduction of ball-milled NaAlBr₄.

Keywords NaAlBr₄ · Solid electrolyte · Na⁺ conductor

Introduction

Li-ion batteries (LIBs) are widely utilised as power sources in portable devices owing to their high energy density. However, because of limited Li reserves, Na-ion batteries (SIBs), in which the Li ions in LIBs are replaced with Na ions, have spurred interest [1]. The reduction voltages of Li⁺ and Na⁺ are lower than the onset voltage of H₂ release in aqueous solutions, necessitating the use of non-aqueous organic electrolytes in conventional LIBs and SIBs; this poses a common safety concern. To ensure battery safety, non-flammable ceramics that facilitate the smooth migration of Li⁺ or Na⁺ have been developed and used as solid electrolytes for all-solid-state Li or Na batteries.

High Li⁺ and Na⁺ conductivities have been reported in oxides [2, 3], sulphides [4–6], and Li/Na complex hydrides [7, 8]. In 2018, Asano et al. reported high Li⁺ conductivities for Li₃YCl₆ and Li₃YBr₆ and stable charge–discharge cycles of their corresponding cells [9]. Further, numerous recent studies have reported the high ionic conductivity of alkali ternary metal halides, with Li⁺ conductivities of the order of 10^{-3} S/cm at room temperature (RT) (e.g., Li₃HoBr₆ [10], Li₃InCl₆ [11], and Li₃ScCl₆ [12]). In addition to chlorides and bromides, Li metal fluorides have emerged as candidates for solid electrolyte development owing to their high electrochemical stability and humidity tolerance [13, 14]. Until 2018, metal halides were not considered potential candidates for solid electrolytes owing to their insufficient conductivity at RT [9]. Although the ionic conductivities of a series of Li/Na metal halides (e.g., Li₂MCl₄ (*M* = Cr, Co, Fe), Li₂TCl₄ (*T* = Mg, Mn, Fe, Cd)) and Na₂MCl₄ (*M* = Mg, Cr, Cd) have been systematically studied, they have remained below 10^{-6} S/cm at RT [15–18]. Meanwhile, the RT conductivity of LiAlCl₄ at 10^{-6} S/cm [19] has been enhanced by partial Fe substitution [20]; however, its application in solid electrolytes has not been frequently reported [21]. Nevertheless, Li/Na–Al chlorides have recently garnered renewed attention owing

✉ Reona Miyazaki
miyazaki.reona@nitech.ac.jp

¹ Department of Physical Science and Engineering, Graduate School of Engineering, Nagoya Institute of Technology, Aichi 466-8555, Japan

² Department of Advanced Ceramics, Graduate School of Engineering, Nagoya Institute of Technology, Aichi 466-8555, Japan

to their stability towards Li and Na metals, as confirmed by stable Li-plating/stripping cycles in Li/LiAlCl₄/Li cells [22]. Na⁺ conductivity has also been reported for NaAlCl₄ [23]. In 2024, Xu et al. reported that the partial fluorination of NaAlCl₄ enhanced the Na⁺ conductivity [24]. Substituting Cl[−] with larger tetrahedral BH₄[−] improved the structural flexibility of AlCl₄[−] [25]. Transition-metal ternary halides such as LiAlCl₄ and NaAlCl₄ have also garnered interest; however, studies on alkaline Al bromide compounds are limited. Br[−] ions, known as ‘soft’ bases, are deemed more suitable for facilitating high Na⁺ conductivity. NaAlBr₄ shares isostructural similarities with NaAlCl₄, simplifying the fabrication and evaluation processes. However, limited studies have reported the conductivities of NaAlBr₄ solids.

In this study, NaAlBr₄ was prepared by ball milling, and its Na⁺ conduction properties were investigated. Molecular dynamics (MD) simulations were employed to simulate the diffusion pathway of Na⁺ in NaAlBr₄.

Experimental

Sample preparation

All samples were prepared and stored in an Ar-filled glove box with ~1 ppm O₂. NaAlBr₄ was synthesised through ball milling (PULVERISETTE 7, Fritsch Co., Ltd). NaBr (99.99%) and AlBr₃ (98%) were purchased from Sigma-Aldrich Co. and Fujifilm Wako Co., respectively. Equimolar amounts of NaBr and AlBr₃ were combined using an alumina mortar and sealed in a chrome steel vessel containing 10 balls (10 mm in diameter). Ball milling was conducted following a similar procedure described in a previous study [26]. Ball-milling cycles were conducted at 400 rpm for 15 min with intervals of 5 min between each cycle. The cycles were repeated 20 times for a total milling duration of 5 h.

X-ray diffraction (XRD) measurement

To identify the crystalline phase of the samples, XRD measurements were conducted employing a CuK α radiation source (MiniFlex, Rigaku CO.). The powdered sample was uniformly mounted on a silica glass plate and sealed using an Al dome and Kapton tape.

Scanning electron microscopy (SEM) observation

The fracture surfaces of the pellets were examined by SEM (JCM-7000, JEOL). Pellets were gently crushed in a glove box using a mortar and pestle. A fragment of the pellet was affixed to a holder using C tape. To mitigate

sample degradation due to hydration and oxidation, air exposure during sample transfer to the SEM apparatus was minimised.

Electrochemical measurement

The powder samples underwent uniaxial pressing at approximately 270 MPa for 10 s using a polyether ether ketone (PEEK) mould (inner diameter of 10 mm). The resulting pellets were not subjected to heating. Subsequently, the pellets were fixed in the mould, and stainless steel (SS) discs (0.2 mm in thickness) were placed on both sides of the pellets as blocking electrodes. The PEEK mould containing the pellets and two SS discs was sealed in the hermetic cell and tightened at 6 Nm. To prevent leakage, the lateral side of the cell was sealed with Kapton tape. The conductivity of the sample was measured using the alternating current impedance method (Solartron 1260) over a frequency range of 1 MHz to 1 Hz. Impedance measurements were conducted within a temperature range of 30–80 °C at intervals of 10 °C. The cell temperature was monitored using a thermocouple positioned directly beneath the cell. A Na/Sn film cell was assembled using NaAlBr₄. The fabrication process was the same as that for the impedance measurement, except for the use of Sn film and Na tip electrodes. The Sn film (approximately 100 nm in thickness) was deposited via vapour deposition onto the SS disc substrate (ϕ 10 mm, 0.2 mm in thickness). Charge–discharge measurements were conducted at 60 °C under a current density of 12.7 μ A/cm² (580, Scribner Co.) within a voltage range of 0.04–1.50 V vs. Na⁺/Na.

MD simulation

The Preferred Potential (PFP) pretrained neural network potential (NNP) integrated into Matlantis software [27] was employed to calculate the energy and forces acting on the ions within a given structure. PFP was trained on approximately 10⁷ first-principles calculations. Structural relaxation and MD simulations were conducted under a canonical ensemble (NVT) using the Atomic Simulation Environment (ASE) [27]. NVT MD simulations were performed for 1 ns with a time step of 1 fs after structural relaxation, unless mentioned otherwise. The validity of the NNP was confirmed by comparing the energies or forces obtained via density functional theory (DFT) calculations. NNP–MD simulations were performed at 373 K for Na₁₆Al₁₆Br₆₄ without point defects, Na₁₉Al₁₅Br₆₄ with interstitial Na⁺ and Na⁺ at the Al site, and Na₁₅Al₁₆Br₆₄ with Na⁺ and Br⁺ vacancies. Plane-wave basis and projector-augmented wave methods [28] were implemented in the Vienna ab initio simulation

package (VASP) [29–31]. The Perdew–Burke–Ernzerhof generalised gradient approximation for solids [32, 33] was applied to approximate the electron–electron interactions. The accuracy of the NNP calculation was verified by comparison with the DFT calculations. Small superstructures of (a) 50 stoichiometric models of $\text{Na}_{16}\text{Al}_{16}\text{Br}_{64}$, (b) 60 Na vacancy models of $\text{Na}_{14}\text{Al}_{16}\text{Br}_{62}$, and (c) 60 interstitial Na models of $\text{Na}_{19}\text{Al}_{15}\text{Br}_{64}$ were generated. The lattice parameters and positions of the consisting atoms were randomly displaced within trace amounts. The prepared structural models were calculated, and the energies were compared with the DFT and NNP (without structural relaxation) results. Figure S1 shows the diagnostic plot with the coefficient of determination R^2 are also close to 1, confirming the potential validity of the NNP. Accordingly, PFP, which was pretrained on the NNP, can reproduce the first-principles DFT outputs for NaAlBr_4 and its relatives well.

Results

XRD results

Figure 1 shows the XRD patterns of Li^+ -doped NaAlBr_4 prepared by ball milling for 5 h. All peaks were indexed to the Pnma space group. NaAlBr_4 has an orthorhombic crystal structure, which is the isostructure of NaAlCl_4 [23, 34]. In the NaAlBr_4 lattice, Na^+ ions are at the centre of the triangular prism (Fig. 1b–d). The nearest-neighbour distance of Na^+ is relatively large (5.7 Å), indicating that continuous Na^+ transport is difficult to achieve without lattice defects.

Electrical conductivity and charge–discharge results

The Arrhenius plot of the conductivities of NaAlBr_4 is depicted in Fig. 2. The conductivity of NaAlBr_4 at 30 °C was 1.2×10^{-5} S/cm, surpassing that of isostructural NaAlCl_4 [23]. The activation energy, calculated from

Fig. 1 **a** XRD pattern of ball-milled NaAlBr_4 . The theoretical pattern for NaAlBr_4 is presented in red. **b–d** Crystal structures of NaAlBr_4 projected into the *ac*, *ab*, and *bc* planes. Sky blue, brown, and yellow balls represent Al, Br, and Na atoms, respectively. NaBr_6 prisms are denoted in yellow

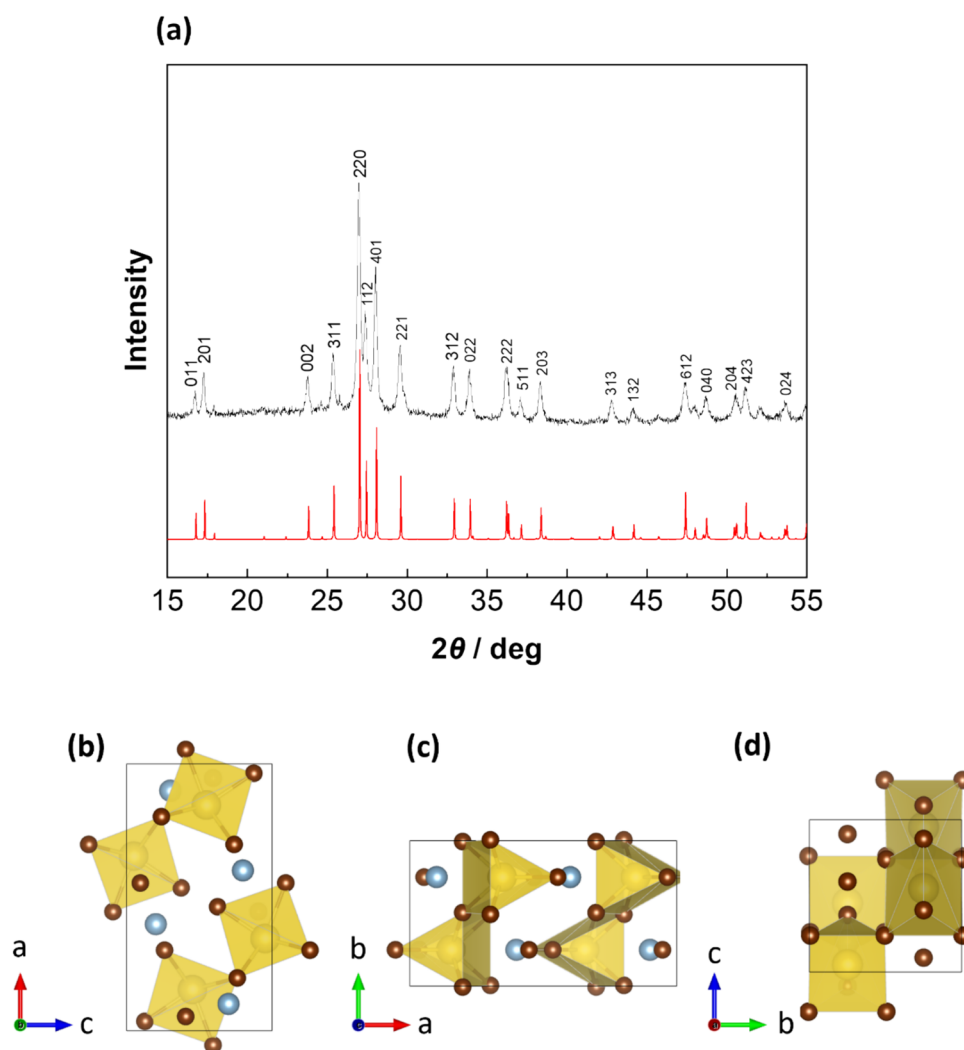
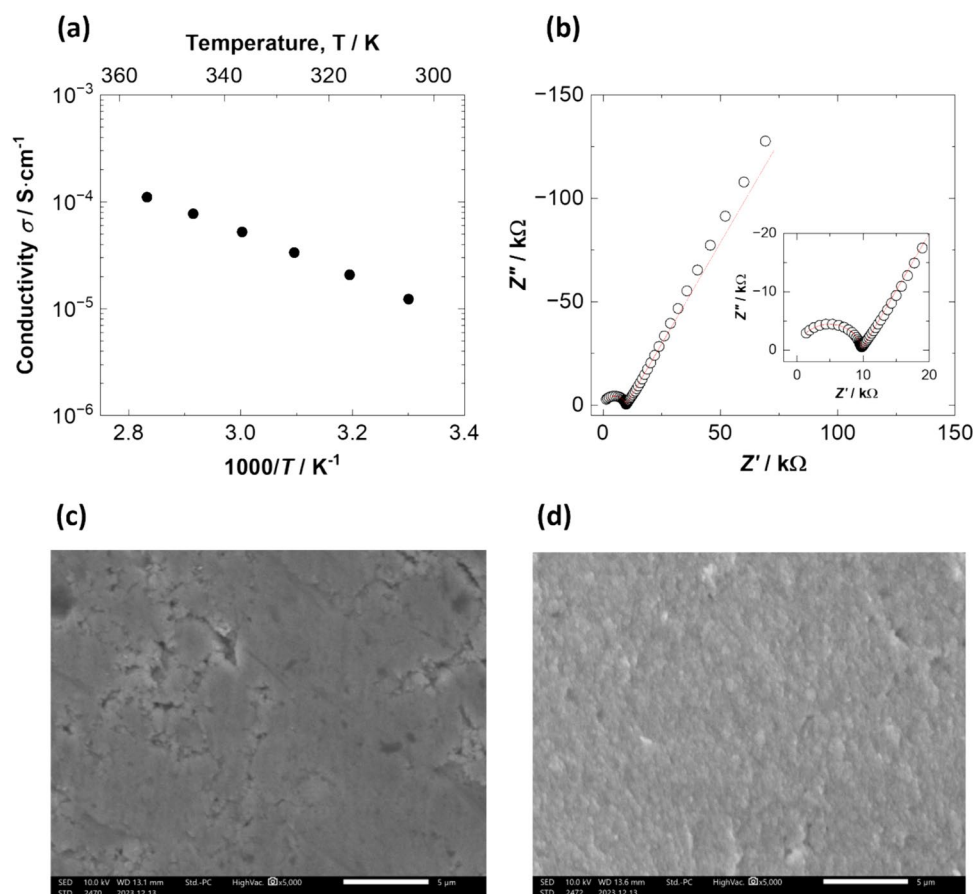


Fig. 2 **a** Arrhenius plot of the conductivity of NaAlBr₄. **b**: Nyquist plots for NaAlBr₄ at 30 °C. Fitting results are plotted by red lines. SEM image of the **d** surface and **e** cross-section of a NaAlBr₄ pellet



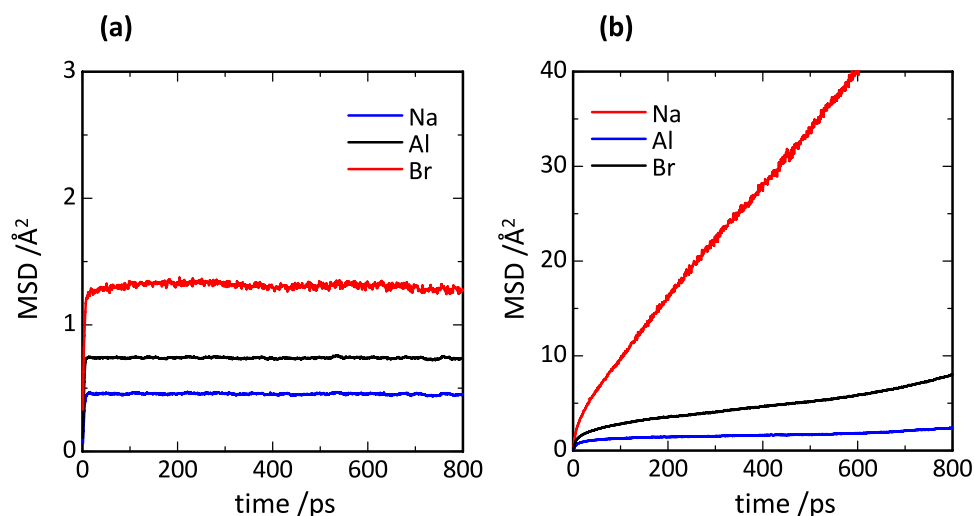
the Arrhenius plot slope, is 0.43 eV. The Nyquist plot at 30 °C can be fitted by a single parallel component of R and constant phase element (Fig. 2b) with an estimated value of 1.22×10^{-10} F, indicating that the grain boundary resistance is slightly mixed with that of the bulk component. The SEM images in Fig. 2c and d show the particle coarsening in the pellet. Despite the presence of pores in the uniaxially pressed pellet, NaAlBr₄ exhibits excellent powder-pressing ability.

For the Na/Sn cell, the initial charging capacity is maintained at 24 mAh/g (Fig. S2). Na metal has low stability, and the continuous charge–discharge of Sn was not observed, unlike in LiAlCl₄ and NaAlCl₄ [22, 23]. The Coulombic efficiencies were less than 84% during cycling, indicating a parasitic reduction reaction. Previous reports suggested the decomposition of NaAlCl₄ to NaCl and Al via $\text{NaAlCl}_4 + 3\text{Na}^+ + 3\text{e}^- \rightarrow \text{Al} + 4\text{NaCl}$ below 1.494 V (vs. Na⁺/Na) [23]. The reduction of AlBr₄⁻ is expected to occur smoothly at higher voltages than that of AlCl₄⁻, following the hard and soft acid–base rule. The soft Br⁻ base may not stably bond to the hard acidic Al³⁺. This trend has been quantitatively observed at the melting temperatures of AlCl₃ (200 °C [35]) and AlBr₃ (97.45 °C [36]).

NNP–MD simulation

From a crystallographic perspective, Na vacancies coupled with interstitial Na⁺ (Frenkel pair) are typically required for Na⁺ transport in bulk NaAlBr₄, given the large nearest-neighbour distance of Na⁺ (> 5.7 Å). In a previous report by Park et al., the Na⁺ Frenkel pair was identified by the Rietveld analysis of ball-milled NaAlCl₄, and Na⁺ transport via interstitial prism sites was suggested [23]. Because of the identical crystal structures of NaAlBr₄ and NaAlCl₄, similar conduction mechanisms are expected for NaAlBr₄. However, contrary to previous assumptions, our NNP–MD calculation results indicated that the presence of Na⁺ Frenkel pair is unlikely in NaAlBr₄. The crystal structure obtained from the Materials Project (mp-1210155) was employed for structural modelling. A 4 × 4 × 2 supercell was created for the original structure (768 atoms in total), followed by structural relaxation and optimisation of the lattice constants. NNP–MD simulations were then conducted over the temperature range 298–673 K for 1 ns, and diffusion coefficients were determined using the mean square displacement (MSD). Representative MSD plots are presented in Fig. 3a and b for stoichiometric NaAlBr₄. The MSD values for all

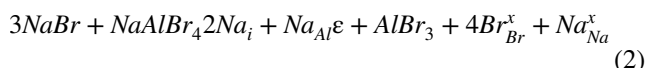
Fig. 3 Mean square displacement (MSD) for Na, Al, and Br ions in stoichiometric NaAlBr₄ at **a** 200 °C and **b** 300 °C



elements remained below $1.5 \text{ \AA}^2$ at temperatures below 200 °C, indicating that Na, Al, and Br were bound to their respective sites and underwent thermal vibration. The MSD values for Na ions were higher than those of the other ions, suggesting thermal vibration in large spaces where Na is prismatically coordinated to Br (NaBr₆). In the MSD profiles above 300 °C, a continuous monotonic increase in MSD was observed for all elements, indicating the melting of the compounds. Melted materials determined on the MSD profiles of Al and Br were excluded from further analysis. In defect-free NaAlBr₄, ion diffusion was not observed, suggesting two models for the defects. The first model is the formation of Na and Br vacancies accompanying the precipitation of NaBr.



The other model is the generation of interstitial Na ions accompanying the precipitation of AlBr₃ in the presence of excess NaBr.



To verify defect generation, five structures with randomly placed point defects were created in the superstructure and relaxed using the NNP. The structures for NaBr and AlBr₃ were obtained from the literature [37, 38]. From the obtained results, the defect formation energies per Na vacancy and interstitial Na were 0.88 and 0.73 eV, respectively. Subsequent NNP–MD calculations were conducted for models with Na vacancies and interstitial Na⁺. Both models were created to have 4%–5% Na vacancies or interstitial Na (Na₁₂₄Al₁₂₈Br₅₀₈ and Na₁₃₄Al₁₂₆Br₅₁₂ for the Na vacancy and interstitial Na models, respectively, versus the stoichiometric composition of Na₁₂₈Al₁₂₈Br₅₁₂). Compositions with low defect concentrations were chosen to minimise the interactions between defects. Additionally, a certain number of

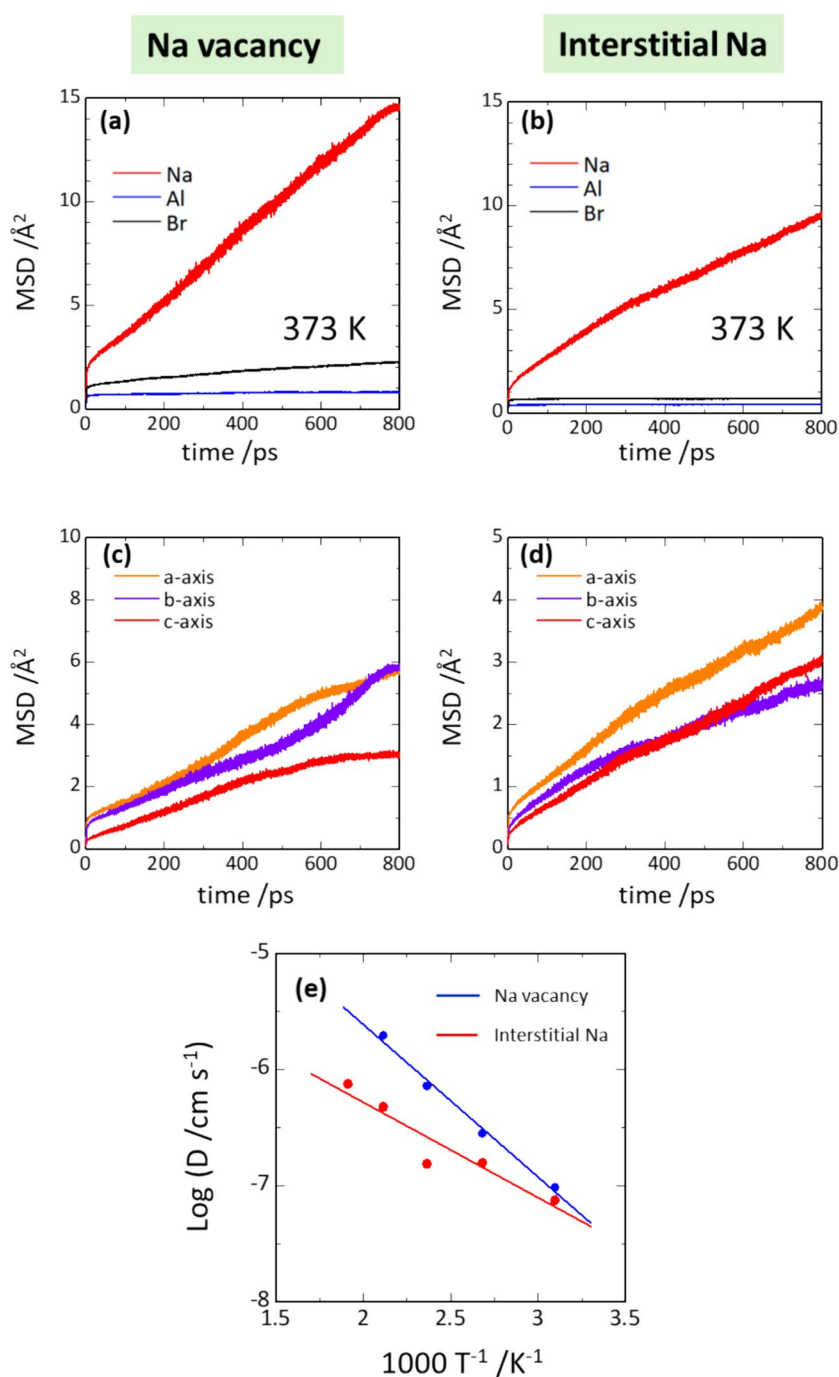
defects was introduced to obtain sufficient statistical data on the hopping phenomenon of Na⁺ by MD calculations. The arrangement of point defects was randomly organised into five models, from which the most stable ones were selected.

Figure 4a and b show the representative MSD profiles for the compositions containing defects. The MSD values of Na⁺ exhibit a monotonic linear increase with the simulation time, indicating Na⁺ diffusion. Additionally, as shown in Fig. 4c and d, the MSD values of Na projected along the a, b, and c axes exhibit similar changes, confirming the three-dimensional conduction. The MSD-derived diffusion coefficients for Na⁺ are plotted in an Arrhenius plot in Fig. 4e. A slight Br diffusion is observed above 373 K in the system with introduced Na and Br vacancies, whereas no diffusion behaviour is detected for the Al ions. The diffusion coefficients are 2–4 times larger in the vacancy model compared to those in the interstitial Na⁺ model at temperatures above 323 K. However, the migration energy differs, being 0.16 eV for interstitial Na and 0.26 eV for Na vacancies, resulting in a higher diffusion coefficient for interstitial Na below 298 K.

Figure 5 illustrates the population density distribution of Na⁺, demonstrating the large thermal vibrations along the ab direction. In the interstitial Na and Na vacancy models, diffusion primarily occurs along the ab plane, traversing through the interstitial regions between Na sites. Additionally, the diffusion paths exhibit a zigzag pattern along the c-axis. Despite small differences in the migration paths between the Na vacancy and interstitial Na models, they share common diffusion pathways.

Figure 6 shows the radial distribution functions (RDFs) for Na–Br and Na–Al interactions averaged over MD simulations at 373 and 473 K. As indicated in Fig. 5, prominent features of the RDF profiles are in agreement with three models, i.e., perfect NaAlBr₄, interstitial Na⁺, and Na⁺ vacancy models. Thus, the structural skeleton of the NaAlBr₄ structure was maintained during MD simulations. In detail, for interactions

Fig. 4 Mean square displacement (MSD) of Na, Al, and Br ions in the **a** Na vacancy (Eq. 1) and **b** interstitial Na (Eq. 2) models. **c, d** MSD for projection onto the a, b, and c axes for the Na vacancy and interstitial Na models. **e** Arrhenius plots for the Na-ion diffusivity of the Na vacancy and interstitial Na models



beyond the second nearest neighbour, a slight difference is observed in the Na^+ vacancy model compared to the other two models, resulting in the broadening of RDF peaks. No clear differences are noted in the RDF profiles obtained at 373 and 473 K. Hence, the peak broadening can be ascribed to the defect-induced structure distortion rather than amorphization. As the Na^+ vacancy model includes the Br vacancy, which is the largest ion in NaAlBr_4 , we reason that the Br vacancy can cause structural distortion in the lattice. Meanwhile, Na occupation at the Al sites does not cause significant differences

compared to the perfect lattice. Thus, the lower migration energy estimated for the interstitial Na^+ model is attributed to the negligible distortion in the crystal structure.

Discussion

The MD simulation results indicate the smooth migration of Na^+ in the NaAlBr_4 lattice with Na vacancies and/or interstitial Na^+ . However, the high formation energies of

Fig. 5 **a** Unit cell structure of NaAlBr_4 and calculated Na population density isosurfaces for **b** Na vacancy and **c** interstitial Na models. The population density increases from the red to blue regions in the coloured map

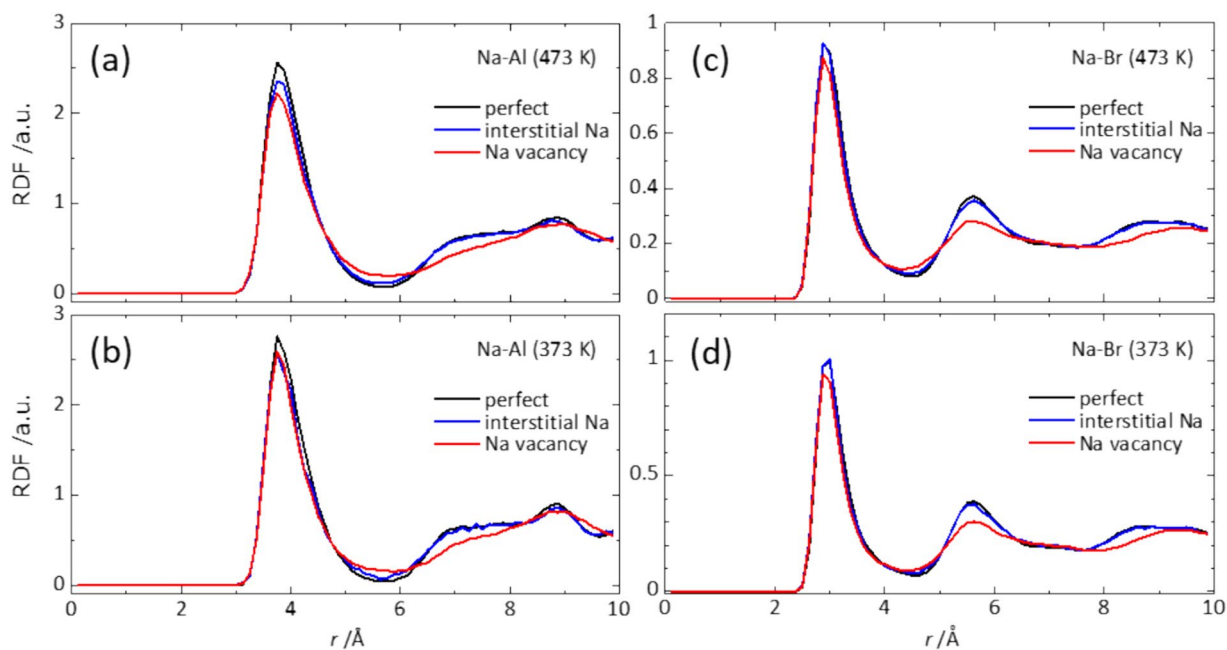
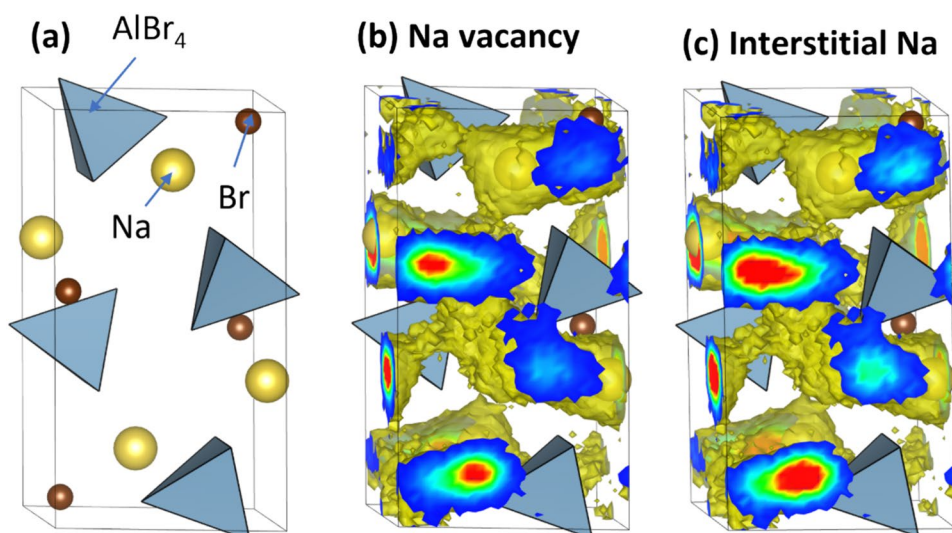


Fig. 6 Radial distribution functions (RDFs) for Na–Al and Na–Br interactions in three calculation models. **a** Na–Al interaction at 473 K. **b** Na–Al interaction at 373 K. **c** Na–Br interaction at 473 K. **d** Na–Br interaction at 373 K

these defects (0.88 and 0.73 eV for the Na^+ vacancy and interstitial, respectively) suggest their unlikely presence in stoichiometric NaAlBr_4 . As off-stoichiometries were not included in this study for NaAlBr_4 , Na^+ conduction was not expected. Nonetheless, the observed ionic conductivity of 10^{-5} S/cm (at 30 °C) suggests that the Na^+ conduction mechanisms may not fully account for it. Considering the activation energy for migration, the vacancy (0.26 eV) and interstitial (0.16 eV) conduction mechanisms cannot explain the experimental results (0.43 eV). The ball-milling preparation method likely induces excess Na^+ and/or Na^+ vacancies

at the particle interphase [39–41]. As the defect formation energies for Na^+ and Na^+ interstitials are similar, both types of defects are considered to coexist. Na^+ ions may preferentially migrate through the particle surface via transiently introduced defects, rather than through bulk NaAlBr_4 , partially explaining the discrepancy in the migration energy between the experimental and simulation results.

Figure 7 illustrates the degradation in the conductivity after repeated heating/cooling cycles between 120 and 30 °C, with the activation energy increasing to 0.49 eV. These results qualitatively indicate the disappearance of

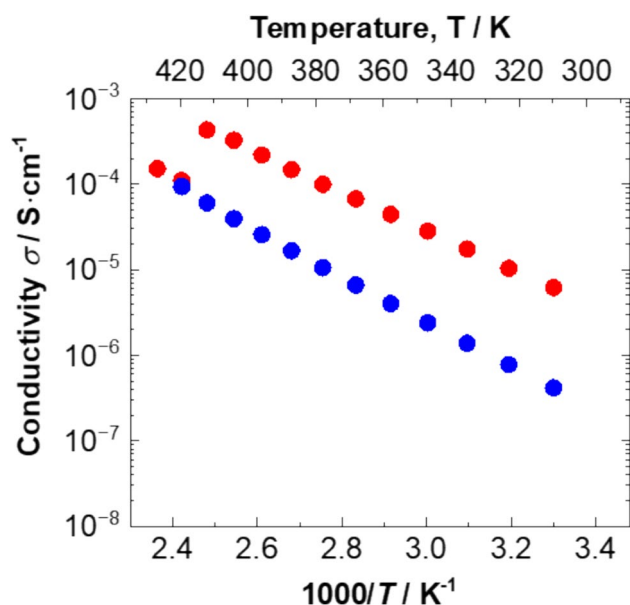


Fig. 7 Arrhenius plot of the conductivity of NaAlBr₄ measured at 30–120 °C. The data obtained from the heating and cooling cycles are presented in red and blue, respectively

defects after annealing, implying a further decrease in the NaAlBr₄ conductivity at higher temperatures, consistent with the tendencies observed for NaAlCl₄ [23]. However, conductivity measurements of up to 150 °C led to NaAlBr₄ decomposition (Fig. S3). Hence, the high-temperature annealing behaviour was not investigated. As intentional doping was not conducted, the NaAlBr₄ conductivity could be enhanced by introducing Na⁺ vacancies and/or interstitial Na⁺.

Conclusions

In this study, the ion conductivity of NaAlBr₄ was investigated for the first time. NaAlBr₄ was fabricated by ball milling. The conductivity of NaAlBr₄ was measured as 1.2×10^{-5} S/cm at RT, surpassing that of isostructural NaAlCl₄. The charge–discharge properties of the Na/Sn cell with NaAlBr₄ exhibited instability, with the initial charge capacity of 24 mAh/g significantly decreasing with cycling, accompanied by Coulombic efficiencies below 84%, denoting a parasitic reduction reaction. NNP–MD simulation results indicated that Na⁺ conduction was not expected in stoichiometric NaAlBr₄, with the defect formation energies estimated as 0.88 and 0.73 eV for the Na⁺ vacancy and interstitial, respectively. These high values suggest that the presence of Schottky or Frenkel pairs is unlikely. The activation energy for Na⁺ conduction derived from the Arrhenius plot was 0.43 eV. Based on the NNP–MD simulation, the migration energies of Na vacancies and interstitials were 0.26 and

0.16 eV, respectively. This discrepancy suggests different conduction mechanisms. The migration energies calculated by the MD simulation were similar to those for the bulk, unlike the case with the defect-enriched particle surface as a transient Na⁺ conduction pathway. As no dopant was intentionally used, further optimisation of the impurity concentration may enhance the Na⁺ conductivity in NaAlBr₄. Thus, this study suggests the potential of NaAlBr₄ as a solid electrolyte. Further explorations of cost-effective Na–Al halides can realise the discovery of novel solid electrolytes with high Na⁺ conductivities and powder-pressing abilities.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10008-024-06086-z>.

Acknowledgements The authors acknowledge financial support from JSPS KAKENHI (grant numbers JP23K23447 and JP22K04871) and NAGAI Foundation for Science and Technology.

Author contribution Reona Miyazaki: conceptualisation, methodology, investigation, data analysis, writing of the original draft, review and editing, funding acquisition, and validation. Masanobu Nakayama: MD simulation, first-principles calculation, writing of the original draft. Takehiko Hihara: supervision and project administration. All authors participated in manuscript preparation.

Funding Open Access funding provided by Nagoya Institute of Technology.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Gao Y, Zhang H, Peng J, Li L, Xiao Y, Li L, Liu Y, Qiao Y, Chou SL (2024) A 30-year overview of sodium-ion batteries. *Carbon Energy* 6:e464
- Thangadurai T, Kaack H, Weppner WJF (2003) Novel fast lithium ion conduction in garnet-type Li₃La₃M₂O₁₂. *J Am Ceram Soc* 86:437–440
- Cussen EJ (2010) Structure and ionic conductivity in lithium garnets. *J Mater Chem* 20:5167–5173
- Kamaya N, Homma K, Yamakawa Y, Hirayama M, Kanno R, Yonemura M, Kamiyama T, Kato Y, Hama S, Kawamoto K, Mitsui A (2011) A lithium superionic conductor. *Nat Mater* 10:682–686

5. Mizuno F, Hayashi A, Tadanaga K, Tatsumisago M (2006) High lithium ion conducting glass-ceramics in the system $\text{Li}_2\text{S}-\text{P}_2\text{S}_5$. *Solid State Ion* 177:2721–2725
6. Mizuno F, Hayashi A, Tadanaga K, Tatsumisago M (2005) New, highly ion-conductive crystals precipitated from $\text{Li}_2\text{S}-\text{P}_2\text{S}_5$ glasses. *Adv Mater* 17:918–921
7. Dimitrievska M, Shea P, Kweon KE, Bercx M, Varley JB, Tang WS, Skripov AV, Stavila V, Udovic TJ, Wood BC (2018) Carbon incorporation and anion dynamics as synergistic drivers for ultrafast diffusion in superionic $\text{LiCB}_{11}\text{H}_{12}$ and $\text{NaCB}_{11}\text{H}_{12}$. *Adv Energy Mater* 8:1703422
8. Udovic TJ, Matsuo M, Tang WS, Wu H, Stavila V, Solonin AV, Skoryunov RV, Babanova OA, Skripov AV, Rush JJ, Unemoto A, Takamura H, Orimo S (2014) Exceptional superionic conductivity in disordered sodium decahydro-closo-decaborate. *Adv Mater* 26:7622–7626
9. Asano T, Sakai A, Ouchi S, Sakaida M, Miyazaki A, Hasegawa S (2018) Solid halide electrolytes with high lithium-ion conductivity for application in 4 V class bulk-type all-solid-state batteries. *Adv Mater* 30:1803075
10. Shi X, Zeng Z, Sun M, Huang B, Zhang H, Luo W, Huang Y, Du Y, Yan C (2021) Fast Li-ion conductor of Li_3HoBr_6 for stable all-solid-state lithium-sulfur battery. *Nano Lett* 21:9325–9331
11. Li X, Liang J, Luo J, Norouzi Banis M, Wang C, Li W, Deng S, Yu C, Zhao F, Hu Y, Sham T-K, Zhang L, Zhao S, Lu S, Huang H, Li R, Adair KR, Sun X (2019) Air-stable Li_3InCl_6 electrolyte with high voltage compatibility for all-solid-state batteries. *Energy Environ Sci* 12:2665–2671
12. Liang J, Li X, Wang S, Adair KR, Li W, Zhao Y, Wang C, Hu Y, Zhang L, Zhao S, Lu S, Huang H, Li R, Mo Y, Sun X (2020) Site-occupation-tuned superionic $\text{Li}_x\text{ScCl}_{3+x}$ halide solid electrolytes for all-solid-state batteries. *J Am Chem Soc* 142:7012–7022
13. Miyazaki R, Yamaguchi G, Yagi E, Yoshida T, Tomita T (2024) Enhancement of the Li^+ conductivity of Li_3AlF_6 for stable all-solid-state lithium-ion batteries. *ACS Appl Energy Mater* 7:2533–2541
14. Miyazaki R, Ozaki S, Yagi E, Sato Y, Kobayashi Y, Yoshida T, Katsuda Y (2022) All-solid-state lithium-ion batteries using the Li_3AlF_6 -based composite as the solid electrolyte. *ACS Appl Energy Mater* 5:15365–15372
15. Kanno R, Takeda Y, Yamamoto O (1981) Ionic conductivity of solid lithium ion conductors with the spinel structure— Li_2MCl_4 . *Mater Res Bull* 16:999–1005
16. Kanno R, Takeda Y, Yamamoto O, Cros C, Gang W, Hagenmuller P (1986) Ionic conductivity and phase transition of the bromide spinels. *J Electrochem Soc* 133:1052
17. Kanno R, Takeda Y, Yamamoto O, Cros WG, Hagenmuller P (1986) Phase transition of the solid lithium ion conductor with the spinel structure $\text{Li}_{2-2x}\text{M}_{1+x}\text{Cl}_4$ ($\text{M}=\text{Mg}, \text{Mn}$). *Solid State Ion* 20:99–103
18. Kanno R, Takeda Y, Murata K, Yamamoto O (1990) Crystal structure of double chlorides, Na_2MCl_4 ($\text{M}=\text{Mg}, \text{Cr}, \text{Cd}$) – Correlation with ionic conductivity. *Solid State Ion* 39:233–244
19. Weppner W, Huggins RA (1976) Ionic conductivity of alkali metal chloroaluminates. *Phys Lett* 58A:245–248
20. Spiesser M, Palvadeau P, Guillot C, Cerisier J (1983) Physical properties of $\text{LiAl}_x\text{Fe}_{1-x}\text{Cl}_4$ ionic conductors. *Solid State Ion* 9–10:103–106
21. Plichta EJ, Behl WK, Yujic D, Chang WHS, Schleich DM (1992) The rechargeable $\text{Li}_x\text{TiS}_2/\text{LiAlCl}_4/\text{Li}_{1-x}\text{CoO}_2$ solid-state cell. *J Electrochem Soc* 139:1509–1513
22. Tanibata N, Takimoto S, Nakano K, Takeda H, Nakayama M, Sumi H (2020) Metastable chloride solid electrolyte with high formability for rechargeable all-solid-state lithium metal batteries. *ACS Mater Lett* 2:880–886
23. Park J, Son JP, Ko W, Kim J-S, Choi Y, Kim H, Kwak H, Seo D-H, Kim J, Jung YS (2022) NaAlCl_4 : New halide solid electrolyte for 3 V stable cost-effective all-solid-state Na-ion batteries. *ACS Energy Lett* 7:3293–3301
24. Xu X, Li Y, Wang X, Tang Y, Gao Q, Tang L (2024) Effect of lattice fluoride and borohydride on the electrochemical performances of NaAlCl_4 solid electrolyte. *J Solid State Electrochem* 28. <https://doi.org/10.1007/s10008-024-05838-1>
25. Lindemann I, Ferrer RD, Dunsch L, Cerny R, Hagemann H, D'Anna V, Filinchuk Y, Schultz L, Gutfleisch O (2011) Novel sodium aluminium borohydride containing the complex anion $[\text{Al}(\text{BH}_4)_2\text{Cl}]^{4-}$. *Faraday Discuss* 151:231–242
26. Miyazaki R, Kurihara D, Hihara T (2016) Li^+ ionic conduction properties on NaI doped with a small amount of LiBH_4 . *J Solid State Electrochem* 20:2759–2764
27. Takamoto S, Shinagawa C, Motoki D, Nakago K, Li W, Kurata I, Watanabe T, Yayama Y, Iriguchi H, Asano Y, Onodera T, Ishii T, Kudo T, Ono H, Sawada R, Ishitani R, Ong M, Yamaguchi T, Kataoka T, Hayashi A, Charoenphakdee N, Ibuka T (2022) Towards universal neural network potential for material discovery applicable to arbitrary combination of 45 elements. *Nat Commun* 13:2991
28. Hafner J (2008) Ab-initio simulations of materials using VASP: density-functional theory and beyond. *J Comput Chem* 29:2044–2078
29. Kresse G, Furthmüller J (1996) Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set. *Comput Mater Sci* 6:15–50
30. Kresse G, Hafner J (1993) Ab initio molecular dynamics for liquid metals. *Phys Rev B Condens Matter* 47:558–561
31. Kresse G, Hafner J (1994) Ab initio molecular-dynamics simulation of the liquid-metal-amorphous-semiconductor transition in germanium. *Phys Rev B Condens Matter* 49:14251–14269
32. Perdew JP, Ruzsinszky A, Csonka GI, Vydrov OA, Scuseria GE, Constantin LA, Zhou X, Burke K (2008) Restoring the density-gradient expansion for exchange in solids and surfaces. *Phys Rev Lett* 100:136406
33. Perdew JP, Burke K, Ernzerhof M (1996) Generalized gradient approximation made simple. *Phys Rev Lett* 77:3865
34. Baenziger NC (1951) The crystal structure of NaAlCl_4 . *Acta Cryst* 4:216–219
35. Berg RW, Hjuler HA, Bjerrum NJ (1984) Phase diagram of the sodium chloride-aluminum chloride system near equimolar composition, with determination of the cryoscopic constant, the enthalpy of melting, and oxide contaminations. *Inorganic Chem* 23:557–565
36. SLEPPY WC (2004) Aluminum halides and aluminum nitrate. *Kirk-Othmer Encycl Chem Technol* 2(26):360
37. Nickels JE, Fineman MA, Wallace WE (1949) X-ray diffraction studies of sodium chloride–sodium bromide solid solutions. *J Phys Colloid Chem* 53:625–628
38. Renes PA, McGillivray CH (1945) The crystal structure of aluminium bromide. *Red Trau Chim Pays-Bas* 64:275
39. Wohlmuth D, Epp V, Stanje B, Welsch AM, Behrens H, Wilkening M, Vyas B (2016) High-energy mechanical treatment boosts ion transport in nanocrystalline $\text{Li}_2\text{B}_4\text{O}_7$. *J Am Ceram Soc* 99:1687–1693
40. Heitjans P, Masoud M, Feldhoff A, Wilkening M (2007) NMR and impedance studies of nanocrystalline and amorphous ion conductors: lithium niobate as a model system. *Faraday Discuss* 134:67–82; 103–118, 415–109
41. Breuer S, Pregartner V, Lunghammer S, Wilkening HMR (2019) Dispersed solid conductors: fast interfacial Li-ion dynamics in nanostructured LiF and $\text{LiF}:\gamma\text{-Al}_2\text{O}_3$ composites. *J Phys Chem C* 123:5222–5230